

A4: Machine Learning - Exam Lab



This exam practice page is designed for the A4: Machine Learning unit of the new IB Computer Science syllabus (first assessment 2027).

Syllabus Links

[A4.1 Machine learning fundamentals eBook](#)

[A4.2 Data preprocessing \(HL Only\) eBook](#)

[A4.3 Machine learning approaches \(HL only\) eBook](#)

[A4.4 Ethical considerations eBook](#)

The Exam Scenario

Scenario: GreenFlow is a technology firm that develops machine learning solutions for commercial farms to optimise crop yields and minimise water waste. They use a variety of sensors and high-resolution cameras to monitor plant health. To deploy these models effectively across various farm environments, ranging from small family plots to massive industrial operations, GreenFlow must carefully consider hardware configurations and the ethics of its automated decision-making processes.

Core Questions (HL and SL)

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 1: Machine Learning Fundamentals (SL & HL)

(a) **Describe** two different approaches to machine learning that GreenFlow might use, providing a specific application for each within the farm scenario. **[4 marks]**

(b) GreenFlow needs to deploy their models on both mobile edge devices (like drones) and central cloud servers. **Describe** the hardware requirements for these two different deployment scenarios. **[4 marks]**

Level 1: Strategy

Command Term: "**Describe**" requires a detailed account.

The Logic: For (a), think about Supervised Learning (labelled images of weeds) versus Unsupervised Learning (clustering different soil types without prior labels). For (b), contrast low-power specialised hardware (Edge) with high-performance scalable infrastructure (Cloud).

Level 2: Mark Scheme

(a) Award **[1]** for identifying an approach (e.g., Supervised Learning) and **[1]** for a relevant application (e.g., classifying images of crops vs. weeds). Repeat for a second approach (e.g., Unsupervised Learning) for soil type clustering.

(b) Award **[1]** for describing edge device requirements (low power, specialised ASICs/FPGAs). Award **[1]** for expansion (e.g., necessity for real-time processing on drones). Award **[1]** for describing cloud server requirements (high-performance GPUs/TPUs, massive storage). Award **[1]** for expansion (e.g., handling large training datasets or complex simulations).

Level 3: Worked Example

(a) GreenFlow could use **Supervised Learning** to train a model on a labelled dataset of images to classify plants as either "healthy" or "diseased". Alternatively, they could use **Unsupervised Learning** (specifically clustering) to analyse sensor data from various fields to group areas with similar irrigation needs without needing pre-labelled soil categories.

(b) For drones (edge devices), hardware must be power-efficient and lightweight, often utilising **ASICs** or **FPGAs** optimised for specific, low-latency inference tasks. In contrast, central cloud servers require advanced infrastructure like **HPC centres** with high-performance **GPUs** or **TPUs** to handle the massive computational complexity and memory usage required for training large-scale models.

Question 2: Ethical Considerations (SL & HL)

(a) **Discuss** the ethical implications of GreenFlow's machine learning models in terms of **algorithmic fairness** and **accountability**. [6 marks]

Level 1: Strategy

Command Term: "**Discuss**" requires a considered and balanced review.

The Logic: Focus on what happens if the training data is biased and who is responsible if an automated irrigation system fails and ruins a crop.

Level 2: Mark Scheme

Award [1-2] for explaining algorithmic fairness (e.g., ensuring models work equally well for different types of crops or farm sizes).

Award [1-2] for the impact of bias in training data (e.g., if data only comes from industrial farms, the AI might not work for small family farms).

Award [1-2] for discussing accountability (e.g., identifying who is responsible—the developer or the farmer—if the AI makes a damaging mistake).

Level 3: Worked Example

Algorithmic fairness is a major concern; if GreenFlow's models are primarily trained on data from large industrial farms, the "optimal" irrigation patterns they suggest might not be effective for small-scale subsistence farms, potentially leading to inequity in the agricultural market. Furthermore, **accountability** is critical; if a reinforcement learning agent incorrectly allocates water, leading to significant crop loss, there must be a clear framework to determine whether the software provider or the end-user is liable. Ensuring **transparency** in how these decisions are made is essential for building trust among farmers.

HL Only: Advanced Machine Learning

Give these questions a try! If you're stuck, check out the 'Strategy' section to help you get started. Once you're finished, use the Mark Scheme and Worked Example to review your progress. If these feel a bit tough right now, head back to the eBook pages to sharpen your understanding first.

Question 3: Data Preprocessing (HL Only)

(a) GreenFlow's sensors often produce "noisy" data due to environmental interference. **Describe** the significance of **data cleaning** in this context and **outline** one technique for handling **outliers**. [4 marks]

(b) **Explain** the importance of **feature selection** in reducing the **curse of dimensionality** for GreenFlow's plant health model. **[3 marks]**

Level 1: Strategy

The Logic: Bad data in = Bad model out. Cleaning fixes this. Feature selection means selecting only the *useful* sensor readings to keep the math from getting too complicated.

Level 2: Mark Scheme

(a) Award **[1]** for the impact of data quality on model performance. Award **[1]** for the significance of cleaning (e.g., removing incorrect data). Award **[1]** for outlining a technique for outliers (e.g., imputation or deletion). Award **[1]** for contextual application.

(b) Award **[1]** for defining feature selection (retaining informative attributes). Award **[1]** for defining the curse of dimensionality (e.g., overfitting, computational complexity). Award **[1]** for explaining the link (fewer features = less complexity/overfitting).

Level 3: Worked Example

(a) **Data cleaning** is significant because the quality of training data directly impacts model performance; incorrect or noisy sensor readings will lead to inaccurate predictions. To handle **outliers** (e.g., a moisture sensor briefly reporting 0% due to a glitch), GreenFlow could use **imputation**, where a predictive model or average value replaces the outlier, ensuring the dataset remains robust for training.

(b) **Feature selection** helps GreenFlow identify and retain only the most informative attributes (e.g., temperature and leaf colour) while discarding irrelevant data. This is crucial to mitigate the **curse of dimensionality**, as having too many features can lead to **overfitting**, where the model learns noise instead of patterns, and significantly increases **computational complexity** and memory usage.

Question 4: Advanced Approaches & Evaluation (HL Only)

(a) GreenFlow is comparing **Decision Trees** and **K-Nearest Neighbours (K-NN)** for classifying soil quality. **Explain** how a Decision Tree would categorise a new soil sample compared to the K-NN approach. **[4 marks]**

(b) GreenFlow prioritises identifying *every* diseased plant, even if it occasionally misidentifies a healthy one as sick. **Explain** why they might focus on **recall** rather than **precision** as an evaluation metric in this specific situation. **[3 marks]**

Level 1: Strategy

The Logic: Decision Trees use "IF-THEN" rules. K-NN looks at the "Neighbours". For (b), Recall is about finding *all* the sick ones. Precision is about being *accurate* when you say one is sick.

Level 2: Mark Scheme

(a) Award **[1]** for explaining Decision Trees, which use learned patterns from labelled data to create a branching logic. Award **[1]** for K-NN using proximity/similarity to existing data points. Award **[1-2]** for distinguishing the mechanical difference (logical rules vs. distance metrics).

(b) Award **[1]** for defining **Recall** (proportion of actual positives correctly identified). Award **[1]** for defining **Precision** (proportion of predicted positives that are correct). Award **[1]** for the justification (missing a disease is more costly than a false alarm).

Level 3: Worked Example

(a) A **Decision Tree** categorises soil by following a logical sequence of branches (e.g., IF pH > 7 AND moisture < 20%) learned from existing data. In contrast, **K-NN** categorises the sample by calculating its "distance" (similarity) to other known soil samples and assigning it the category most common among its closest "neighbours".

(b) GreenFlow focuses on **recall** because it measures the model's ability to find *all* actual instances of diseased plants. In agriculture, the cost of a "false negative" (missing a disease that could spread) is much higher than that of a "false positive" (checking a healthy plant), so maximising recall ensures the safety of the entire crop, even if **precision** (the accuracy of sick-plant alerts) is lower.

How confident do you feel with this topic?

Exam Ready: I answered most questions correctly without looking at the eBook. I can explain these concepts to someone else and understand the "why" behind the mark scheme.

Move on to the next topic.

Practitioner: I understood the general concepts, but I missed a few marks on specific terminology or complex applications. I needed the "Strategy" guide for the harder questions.

Annotate your answers with the Mark Scheme and revisit the eBook sections where you lost marks. You can also use the bookmarking and notes feature on the eBook pages to highlight certain areas.

Novice: I struggled to get started on the questions. I found it difficult to apply the theory to the practice scenarios.

Go back to the eBook and Worked Examples before attempting these questions again. Perhaps use some of the Quizlets to practice the key terms and review the command terms.

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